Question Bank

Type A-Orthogonal Trajectories

Question- Find Orthogonal Trajectories for the following.

1. Find Orthogonal Trajectory of $xy = c$[**Ans:** $x^2 - y^2 = c^2$]
2. Find Orthogonal Trajectory of $y = ax^2$[**Ans:**$\frac{x^2}{(\sqrt{2c})^2} + \frac{y^2}{c^2} = 1$]
3. Find Orthogonal Trajectory of $e^x + e^{-y} = c$ [**Ans:**$\boldsymbol{e^y - e^{-x} = c}$]
4. Find Orthogonal Trajectory of $r = a(1 - cos\theta)$ [**Ans:** $\boldsymbol{r = c(1 + cos\theta)}$]
5. Find Orthogonal Trajectory of $r^2 = a^2 sin2\theta$[**Ans:** $\boldsymbol{r^2 = c^2 \cos(2\theta)}$]

Type B-Newtons Law of Cooling

1. Water at temperature 100^0c cools in 10 minutes to 88^0c in a room of temperature 25^0c . Find the temperature of water after 20 minutes.

   [**Ans:**$\boldsymbol{72.92^0 c}$]

2. A body at temperature 100^0c is placed in a room whose temperature is 20^0c and cools to 60^0c in 5 minutes. Find its temperature after a further interval of 3 minutes.

   [**Ans:**$\boldsymbol{\theta = 46.39^0 c}$]

3. If the temperature of the body drops from 100^0c to 60^0c in one minute when the temperature of surrounding is 20^0c, what will be the temperature of the body at the end of the second minute**?**

[**Ans:** $\boldsymbol{\theta = 40^0\ c}$]

4. A body originally at 80^0c cools down to 60^0c in 20 minutes, the temperature of the air being 40^0c. What will be the temperature of the body after 40 minutes from original?

[**Ans:** $\boldsymbol{\theta = 50^0 c}$]

5. A thermometer is moved from a room where the temperature is **21°C** to an outdoor place where the temperature is **0°C**. After **one minute**, the thermometer reading falls to **10°C.**

[**Ans:** $\boldsymbol{t = 1.934144}$]

How long after the thermometer is taken outside will the reading become **5°C**?

Type-C Simple Electric Circuit

1. A circuit consists of resistance R ohms and a condenser of C farads connected to a constant emf. If $\frac{q}{C}$ is the voltage of the condenser at time t after closing the circuit, show that the voltage at time t is $E\left(1 - e^{-\frac{t}{RC}}\right)$.
2. An emf 200 e^{-5t} is applied to a series electric circuit consisting of 20Ω resistance and 0.01F capacitor. Find the charge and current at any time t, assuming that there is no initial charge on capacitor. [**Ans:** $\boldsymbol{i = 10[1 - 5t]e^{-5t}}$
3. In an electric circuit containing inductance and resistance in series with constant emf E if initial

current is zero. Show that, current builds up half of its theoretical max in $\frac{Llog2}{R} sec.$

4. A resistance of 100Ω, an inductance of $0.5H$ is connected in series with a battery of $20v$. Find the current in a circuit as a function of t.**[Ans $\boldsymbol{i} = \frac{1}{5}(1 - e^{-200t})$]**

5. When switch is closed in circuit containing battery E, resistance R, and inductance L the current I build up at rate given by $L\frac{dI}{dt} + RI = E$. Find I as a function of t. How long will t be, before current has reached up to half its maximum value, if $E = 6V, R = 100\Omega, L = 0.1H$. **[Ans:**$t = 0.00069314$]

Type D-Rectilinear Motion

1. A body of mass m, falling from rest is subjected to the force of gravity and an air-resistance proportional to the square of the velocity (kv^2) if it falls through a distance x and possesses a velocity v at that instant, prove that $\frac{2kx}{m} = log\left(\frac{a^2}{a^2-v^2}\right)$ where $mg = ka^2$.

2. The distance x descended by a parachute satisfies the differential equation $v\frac{dv}{dx} = g\left(1-\frac{v^2}{k^2}\right)$ where v is velocity, k, g, constant. If $v = 0$ and $x = 0$ at time $t = 0$ show that $x = \frac{k^2}{g} logcoh\left(\frac{gt}{k}\right)$.

3. A bullet is fired into a sand tank, its retardation is proportional to the square root of its velocity is $(k\sqrt{v})$. How long will it take come to rest if it enters the sand tank with an initial velocity v_0. **[Ans:**$\boldsymbol{t} = \frac{2}{k}\sqrt{v_0}$]

4. A particle is moving in a straight line with an acceleration $k\left[x + \frac{a^4}{x^3}\right]$ directed towards origin. If it starts from rest at a distance a from the origin, prove that it will arrive at origin at the end of time $\left(\frac{\pi}{4\sqrt{k}}\right)$.

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Type-E Fourier Law of Heat Conduction

1. A pipe $30cm$ in diameter certain steam at 150^0c and it protected with covering $5cm$ thick for which $k = 0.0025$. If the temperature of the outer surface of covering is 40^0c. Find the temperature at point at which is at $16.5cm$ from axis of pipe.

   [Ans: Q=6.003156, T=113.55.]

2. A pipe $10cm$ in diameter contains steam at 100^0c, it is covered with asbestos $5cm$ thick for which $k = 0.006$ and temperature is at 30^0c. Find the amount of heat lost per hour from a meter long pipe.

   [Ans: Q=3.805252]

3. A steam pipe $20cm$ in diameter is protected with covering $6cm$ thick for which the coefficient of thermal conductivity is $k = 0.0003cal/cm$ in steady state. Find the heat loss per hour through a meter length of the pipe if the inner surface of pipe is at 200^0c and outer surface is at 30^0c. Also find

temperature at distance $12cm$ from the center of the pipe.

[Ans: $Q' = 245443.4\ cal, T = 134.02$]

4. A long hallow pipe has an inner diameter $10cm$ and outer diameter $20cm$. The inner surface is kept at 200^0c and outer surface at 50^0c. The thermal conductivity is $k = 0.12$. How much heat loss per minute from a portion of the pipe 20cm long. Find the temperature at distance $x = 7.5cm$ from the center of the pipe.

[Ans: Q=163.0822, T =112].

5. A pipe 20cm in diameter contains steam at 200^0c and is protected with covering 5cm thick for which $\mathrm{k} = 0.0015$. If temperature of outer surface on the covering is at 50^0c. find the temperature half-way through to the covering under steady state condition.

[Ans: Q=3.484886, T= T=117.4490]

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